

YIMIAN LIU

Research Engineering | Machine Learning & Energy System

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EXPERIENCE

Guzman Energy

Denver, CO

Operations Research Engineer (AI & Platform Tech Lead)

Jan 2026 – Present

- Built from scratch and now operate an end-to-end SPP power-market decision platform that cleared **799 GWh** (\$26.14M market notional) in 6+ months with 99.9% submission acceptance. Connects weather and market data to probabilistic forecasts, trader-defined strategies and asset-constrained offers and demand bids; feeds awards, dispatch and settlements back into the next decision cycle.
- Reduced load/generation forecast error (MAPE) by **17%** versus the prior vendor with a production agentic research system using WeatherNext 3/NWP and market data. Designed an autonomous research loop with immutable forecasting artifacts and predictions committed before outcomes are known; an AI Critic selects production models using only past realized evidence, while integrity-cleared research histories guide successor experiments.
- Led design and delivery of **least-cost transmission pathfinding** with capacity constraints across the western network. Integrated minute-level OASIS data from 23 transmission providers (457M+ offering records) and connected optimized routes to transmission reservations and e-tags.
- Built a human-in-the-loop strategy and execution engine that translates p10–p90 forecasts and asset limits into hourly offers for 20+ trader-defined strategies, with independent preflight validation, live-state drift blocking, partial-failure recovery and complete decision lineage.
- Launched an enterprise AI platform serving 51 monthly active users, recording 60,663 events in 5 months. Integrated agent orchestration, BigQuery/MCP tools, user-scoped OAuth 2.0 and role-based access for governed data operations with end-to-end audit trails.
- Lead a three-person engineering team while remaining heavily hands-on in ML research, optimization and production; built an AI-native, research-to-production culture with automated code review and API, testing and CI quality gates.

Software Engineer

Oct 2024 – Dec 2025

- Architected the company's unified internal AI and data platform from scratch using Python/GCP, React/TypeScript and BigQuery; team-owned pipelines account for **1.5B+ production rows**. Built CAISO/SPP battery scheduling, validated market submissions and live position monitoring across market, transmission and settlement workflows.

Elseware Capital | Full Stack Developer (Remote)

Jan 2024 – Sep 2024

- Reduced API response latency by 80% under load tests simulating 1,000+ concurrent users by profiling asynchronous FastAPI endpoints and MySQL queries for a real-estate investment platform.

Ecesis Investment | Frontend Developer Intern (Part-time, Remote)

Jul 2023 – Dec 2023

- Built an AI transcription application; cut streaming latency by 60% through WebSocket optimization and rendering time by 80% through caching and state management.

EDUCATION

Cornell University | M.Eng., Computer Engineering

Aug 2022 – Dec 2023

GPA: 3.95/4.00. Deep Learning (A+), Machine Learning (A), Computer Vision (A+), Distributed Computing Principles.

University of Liverpool | B.Eng. (Honors), Electrical & Electronic Engineering

Sep 2017 – Jul 2021

GPA: 3.97/4.00. Power Systems (A+), Instrumentation/Control, Signals/Systems, Electromagnetism, Field Theory & PDEs.

SELECTED PROJECTS

Transformer & Reinforcement Learning [\[Demo\]](#) Built a Transformer policy using supervised pretraining and PPO for predator-prey simulation; distilled the policy by 40% for <10 ms browser inference.

Distributed Systems [\[Code\]](#) Implemented a sharded Java key/value store using Multi-Paxos and two-phase commit for strongly consistent cross-shard transactions.

TECHNICAL SKILLS

ML & Research: Python, PyTorch, TensorFlow/Keras, Chronos 2, temporal backtesting, time-series foundation models.

Systems & Data: SQL/BigQuery, GCP, FastAPI, Airflow, Java, React/TypeScript, distributed systems, CI/CD.